

EF Core 8.0 Guided Hands-On Exercises — Answer Key

Lab 1: Understanding ORM with a Retail Inventory System

Scenario	You're building an inventory management system for a retail store. The store wants to track products, categories, and stock levels in a SQL Server database.
Objective	Understand what ORM is and how EF Core helps bridge the gap between C# objects and relational tables.

Step 1: What is ORM?

ORM (Object-Relational Mapping) is a technique that lets developers interact with a relational database using object-oriented code instead of writing raw SQL. It acts as a translation layer between C# classes and database tables.

How ORM maps C# classes to tables

C# Concept	Database Concept
Class (e.g. Product)	Table (e.g. Products)
Property (e.g. Name)	Column
Object instance	Row
Navigation property (e.g. Category)	Foreign key relationship

Benefits

- **Productivity** — no need to hand-write repetitive SQL/CRUD code.
- **Maintainability** — schema changes tracked through C# models and migrations, not scattered SQL scripts.
- **Abstraction from SQL** — developers work with familiar objects and LINQ instead of database-specific syntax.

Step 2: EF Core vs EF Framework (EF6)

Feature	EF Core	EF Framework (EF6)
Platform	Cross-platform	Windows-only
Performance	Lightweight, faster	Heavier, more overhead
Async support	Native async/await	Limited
Compiled queries	Supported	Not natively supported

Maturity	Newer, actively evolving	Older, very stable
Flexibility	Modular NuGet packages	Monolithic

Summary: EF Core is the modern, cross-platform choice for new development; EF6 remains relevant mainly for maintaining legacy .NET Framework applications.

Step 3: EF Core 8.0 Features

- **JSON column mapping** — map C# properties/complex types directly to a JSON column.
- **Compiled models** — pre-compile the model at build time to reduce startup time for large models.
- **Interceptors** — hook into and customize EF Core's query/save pipeline (logging, modifying commands, etc.).
- **Better bulk operations** — `ExecuteUpdate()` and `ExecuteDelete()` perform bulk changes directly in the database without loading entities into memory.

Step 4: Create a .NET Console App

```
dotnet new console -n RetailInventory  
cd RetailInventory
```

Scaffolds a new console app named `RetailInventory` (creates `.csproj` + `Program.cs`), then moves into the project folder.

Step 5: Install EF Core Packages

```
dotnet add package Microsoft.EntityFrameworkCore.SqlServer  
dotnet add package Microsoft.EntityFrameworkCore.Design
```

- **SqlServer** — provider that lets EF Core talk to SQL Server (translates LINQ into T-SQL).
- **Design** — design-time tooling required for migrations (`dotnet ef migrations add`, `dotnet ef database update`).

Key Takeaway

ORM eliminates manual SQL by mapping classes to tables and objects to rows. EF Core 8 is the cross-platform, high-performance evolution of EF6, ready to be wired up with a `DbContext`, models, and migrations in the following labs.